Noncoherent Sensing and Interference Nulling for TN–NTN Coexistence in the Upper Mid-Band: Large-Matrix Analysis and Finite-Array Leakage Scaling

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***Abstract*—Terrestrial–satellite downlink coexistence requires spatial protection when receiver identities, uplink waveforms, and instantaneous downlink channels are unavailable. We connect a single-victim generalized likelihood-ratio estimator to large-matrix spatial overlap and residual nulling leakage. Under a common spatial-signature benchmark, separate fixed-array analyses characterize the mean leakage at high sensing SNR and with long observation records. For multiple victims, blind MDL–MUSIC provides anonymous uplink directions and covariance powers. We map these directions to the downlink manifold of a fixed physical array and introduce a nominal frequency-dependent power correction without assuming instantaneous channel reciprocity. The transferred information supports a noncoherent beamformer that balances desired signal gain against a weighted directional-interference penalty at fixed transmit power. Site-specific ray-tracing results quantify the spatial interference suppression at detected NTN receivers and the corresponding TN-link performance.**

***Index Terms*—TN-NTN coexistence, interference nulling, noncoherent detection, channel estimation, FR3, upper mid-band**

I. Introduction

Satellite and non-terrestrial networks (NTN) increasingly share spectrum with terrestrial cellular systems. The upper mid-band (7–24 GHz) offers substantial bandwidth together with more favorable propagation than mmWave bands [1], but terrestrial downlink transmissions can interfere with incumbent NTN receivers [2], [3]. A BS must therefore obtain useful protection information even when receiver identities, locations, and uplink waveforms are unavailable.

Prior work has studied coexistence interference [4], robust beamforming with uncertain channels [5], nulling toward satellites [6], [7], reverse spectrum sharing with joint grouping and beamforming [8], angular radiated-power constraints [9], and satellite selection [10]. These approaches use structured geometry, channel knowledge, or coordination. Passive sensing of unknown NTN uplink waveforms instead provides anonymous spatial information [11], whose usefulness depends on both estimation quality and how it is transferred to the protected link.

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Two distinctions matter for downlink protection. First, when uplink and downlink carriers differ, a common physical direction has different array responses; an uplink steering vector cannot simply be reused as a downlink channel. Cross-band covariance methods exploit shared angular structure [12], [13], but do not supply the unknown instantaneous downlink phases. Second, finite sensing records produce direction errors whose effect on interference depends on the normalized transmit beam. Relating estimation accuracy to leakage therefore requires both the spatial error geometry and the sensing SNR and snapshot budget.

We develop the model in this order. A single-victim benchmark first connects noncoherent estimation to residual leakage. Multi-source sensing then supplies frequency-specific spatial and power inputs to a downlink beamformer. The main components are as follows.

- *Estimation and leakage in a common analytical model:* Profiling the unknown waveform yields a covariance-eigenvector GLRT estimator [14]. Its large-matrix overlap [15] is propagated through an estimated spatial null while retaining the residual error along the desired direction. Separate fixed-array expansions characterize the high-SNR and long-observation regimes.
- *Blind sensing with explicit cross-band transfer:* We retain MDL–MUSIC source enumeration and anonymous covariance-power estimation, reconstruct downlink directions on the fixed physical array, and distinguish nominal frequency-dependent power compensation from complex-channel reciprocity and transmit-power control.
- *Noncoherent downlink beamforming and spatial evaluation:* The transferred directions and nominal power weights define a fixed-power beamforming objective that trades desired gain against interference. Site-specific ray-tracing simulations evaluate the resulting NTN INR and TN-link performance using the physical downlink channels.

The analytical benchmarks and reported simulations evaluate spatial estimation and interference control under their stated sensing and propagation assumptions.

II. System Model and Sensing Information

The BS senses NTN uplink activity at carrier f_{U} and transmits TN downlink at f_{D}, which couples into the protected NTN receiver. The two carriers may differ; frequency separation at

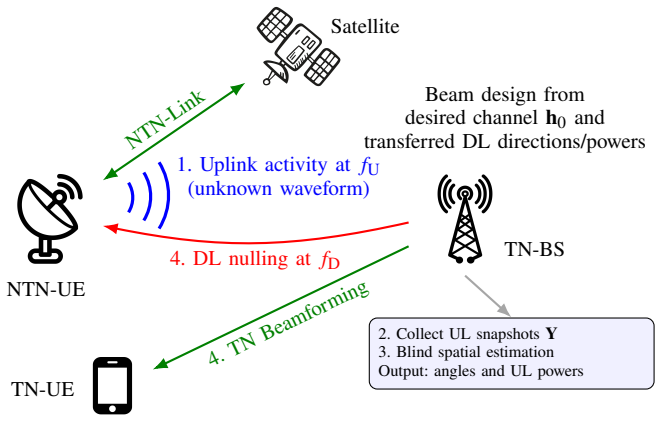


Fig. 1: Noncoherent sensing and downlink interference control. The BS estimates anonymous uplink directions and powers, reconstructs them at the downlink carrier, and combines the transferred information with the desired TN channel for beam design.

the sensing front end is distinct from the spectral coupling that causes interference.

Downlink Interference Model: We consider a terrestrial transmitter (TX) with N_{tx} antennas serving an intended TN user RX_0 while coexisting with victim receivers $\text{RX}_1, \ldots, \text{RX}_N$. Let $\mathbf{w} \in \mathbb{C}^{N_{\text{tx}} \times 1}$ be the unit-norm transmit beamformer, $\boldsymbol{h}_i \in \mathbb{C}^{N_{\text{tx}} \times 1}$ the narrowband channel at f_D from the TX to RX_i, $\mathcal{E}_D$ the downlink symbol energy, and N_0 the receiver noise energy. Then the victim INR is

$$\text{INR}_i = \frac{\mathcal{E}_D |\mathbf{w}^* \boldsymbol{h}_i|^2}{N_0}, \quad i = 1, \ldots, N. \tag{1}$$

For $i = 0$, the same expression gives the desired-link SNR.

Victim Sensing Model: During sensing, the TX collects T array snapshots from victim uplink activity and models the received matrix as

$$\mathbf{Y} = \sum_{i=1}^{K} \mathbf{u}_i \mathbf{b}_i^{\mathsf{T}} + \mathbf{W}, \tag{2}$$

where $\mathbf{Y} \in \mathbb{C}^{N_{\text{tx}} \times T}$, K is the number of victim sources represented in the record, and each unit-norm vector $\mathbf{u}_i$ is the uplink spatial signature of victim i at f_U. The temporal vector $\mathbf{b}_i = [b_i(1), \ldots, b_i(T)]^{\mathsf{T}}$ collects the T uplink samples with $b_i(\tau) = g_i s_i(\tau)$, so it absorbs both the effective sensing coefficient g_i and the transmitted symbol $s_i(\tau)$. Here τ indexes collected snapshots, rather than elapsed time. The noise matrix $\mathbf{W}$ has i.i.d. entries distributed as $\mathcal{CN}(0, \sigma^2)$.

No terminal identity or known pilot is required. The source count K refers to the activity present in the sensing record; a silent uplink does not rule out a vulnerable downlink receiver. The cross-band construction in Section IV transfers the geometry observed in the sensing record.

Listening Windows and Snapshot Budget: The BS collects T snapshots over L_s listening windows. Window ℓ contains n_ℓ snapshots in $\mathbf{Y}_\ell \in \mathbb{C}^{N_{\text{tx}} \times n_\ell}$, giving

$$\mathbf{Y} = [\mathbf{Y}_1, \ldots, \mathbf{Y}_{L_s}], \qquad T = \sum_{\ell=1}^{L_s} n_\ell. \tag{3}$$

Contiguous acquisition has $L_s = 1$; equal-size windows have $n_\ell = n_s$ and $T = L_s n_s$. Figure 2 illustrates one continuous

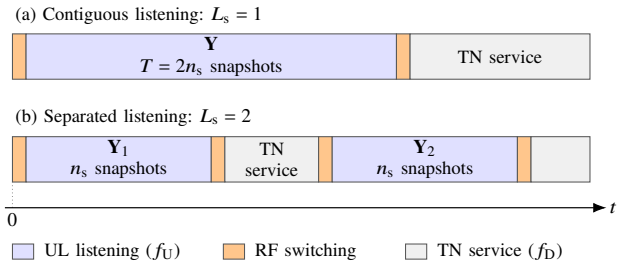


Fig. 2: Shared-RF acquisition of $T = 2n_s$ snapshots: (a) one window; (b) two windows of n_s snapshots each. Both rows use the same illustrative time scale and snapshot rate. Only blue blocks contribute to $\mathbf{Y}$; switching and intervening TN service delay completion. No protocol-slot duration is imposed.

window and two separated windows with the same snapshot budget and effective snapshot rate f_{snap}.

Pooling requires fixed spatial signatures over acquisition and the stated white-noise model across all snapshots. Scalar phase offsets between windows enter $\mathbf{b}_i$, so carrier-phase coherence across windows is unnecessary. Sources may be silent in some windows, but receiver noise is retained. Repartitioning the same record leaves the pooled covariance in (30) unchanged. Section III specifies the additional waveform-energy assumptions for its analytical benchmark.

With a shared RF chain, the BS retunes to f_U for listening and returns to f_D for TN service between windows. The total listening duration is T/f_{snap}; retuning and inter-window gaps add elapsed time without adding snapshots. Section IV-D relates this window model to random uplink activity, coverage, and TN service cost.

III. Single-Victim Noncoherent Estimation and Nulling Leakage

We first specialize (2) to one candidate victim. Profiling the unknown waveform gives a GLRT detector and spatial estimator. We then quantify its large-matrix overlap and connect estimation error to residual leakage under an explicitly defined single-null beam, before extending sensing to multiple victims.

A. GLRT Estimator

For a single candidate victim, (2) reduces to the binary test

$$\begin{aligned} &H_0 : \mathbf{Y} = \mathbf{W}, \\ &H_1 : \mathbf{Y} = \mathbf{u}_1 \mathbf{b}^{\mathsf{T}} + \mathbf{W}, \qquad \|\mathbf{u}_1\|_2 = 1. \end{aligned} \tag{4}$$

With circular Gaussian noise, the GLRT is

$$\Lambda(\mathbf{Y}) = \frac{\max_{\mathbf{u}_1, \mathbf{b}, \sigma^2} p(\mathbf{Y} \mid H_1, \mathbf{u}_1, \mathbf{b}, \sigma^2)}{\max_{\sigma^2} p(\mathbf{Y} \mid H_0, \sigma^2)} \underset{H_0}{\overset{H_1}{\gtrless}} \gamma_G, \tag{5}$$

where γ_G is chosen to meet the target false-alarm probability. For fixed unit-norm $\mathbf{u}_1$, profiling out the unknown waveform gives the least-squares estimate $\widehat{\mathbf{b}}^{\mathsf{T}} = \mathbf{u}_1^{\mathsf{H}} \mathbf{Y}$, so the maximization under H_1 reduces to a Rayleigh-quotient problem in $\mathbf{u}_1$; Appendix A gives the explicit likelihoods and algebra.

$$\min_{\mathbf{b}} \|\mathbf{Y} - \mathbf{u}_1 \mathbf{b}^{\mathsf{T}}\|_F^2 = \|\mathbf{Y}\|_F^2 - \|\mathbf{u}_1^{\mathsf{H}} \mathbf{Y}\|_2^2, \tag{6}$$

This makes the Rayleigh-quotient structure explicit, and the resulting spatial estimate is the principal eigenvector of the sample covariance

$$\mathbf{R} \triangleq \frac{1}{T} \mathbf{Y} \mathbf{Y}^{\mathsf{H}}, \qquad \widehat{\mathbf{u}}_1 = \mathbf{v}_{\max}(\mathbf{R}). \tag{7}$$

Under H_0, the ML noise-variance estimate is

$$\widehat{\sigma}_0^2 = \frac{\|\mathbf{Y}\|_F^2}{N_{\mathrm{tx}}T}, \tag{8}$$

while after profiling out $\mathbf{b}$ and $\mathbf{u}_1$ under H_1,

$$\widehat{\sigma}_1^2 = \frac{\|\mathbf{Y}\|_F^2 - \lambda_{\max}(\mathbf{Y}\mathbf{Y}^{\mathsf{H}})}{N_{\mathrm{tx}}T}. \tag{9}$$

Hence the profiled GLRT statistic becomes

$$\Lambda(\mathbf{Y}) = \left(\frac{\widehat{\sigma}_0^2}{\widehat{\sigma}_1^2}\right)^{N_{\mathrm{tx}}T}. \tag{10}$$

The profiled GLRT is monotone in the normalized largest eigenvalue

$$\xi \triangleq \frac{\lambda_{\max}(\mathbf{R})}{\mathrm{Tr}(\mathbf{R})}, \tag{11}$$

and can therefore be implemented as

$$\lambda_{\max}(\mathbf{R}) \underset{H_0}{\overset{H_1}{\gtrless}} \psi \ \mathrm{Tr}(\mathbf{R}), \tag{12}$$

for an appropriate threshold ψ. This yields both a single-candidate detector and a spatial estimate without knowledge of the transmitted waveform. We next quantify the estimate's overlap with the true signature and its effect on nulling leakage.

B. Large-Matrix Spatial Overlap

Under H_1, write the unknown temporal factor in (4) as $\mathbf{b} = g\mathbf{s}$, where g is the effective uplink sensing coefficient and $\mathbf{s} \in \mathbb{C}^T$ collects the victim uplink symbols. The entries of $\mathbf{W}$ are i.i.d. $\mathcal{CN}(0, \sigma^2)$. The sensing SNR is

$$\gamma_{\mathrm{S}} \triangleq \frac{|g|^2}{\sigma^2}. \tag{13}$$

We quantify the estimation quality through the overlap

$$\eta \triangleq \left|\widehat{\mathbf{u}}_1^{\mathsf{H}}\mathbf{u}_1\right|^2. \tag{14}$$

The GLRT estimate in (7) is the principal left singular vector of $\mathbf{Y}$, so its overlap can be related to the left-factor squared correlation in [15, Th. 5.1, eq. (5.3)]. Let $N_{\mathrm{tx}}, T \to \infty$ with fixed aspect ratio

$$\beta \triangleq \frac{T}{N_{\mathrm{tx}}} \in (0, \infty). \tag{15}$$

Assume $\|\mathbf{s}\|_2^2/T \to 1$ and that the signal factors are independent of the white noise. Normalize the observations by σ and write $\overline{\mathbf{W}} = \mathbf{W}/\sigma$. The model $\mathbf{A} = \mathbf{Y}/\sigma$ then has the form $\mathbf{A} = \mathbf{u}_{\mathrm{r}}\mathbf{s}^{\mathsf{T}} + \sqrt{N_{\mathrm{tx}}}\mathbf{W}_{\mathrm{r}}$ in [15], with $\mathbf{u}_{\mathrm{r}} = (g/\sigma)\mathbf{u}_1$ and $\mathbf{W}_{\mathrm{r}} = \overline{\mathbf{W}}/\sqrt{N_{\mathrm{tx}}}$. Its factor second moments and noise parameter are

$$\begin{aligned} \tau_u &= \gamma_{\mathrm{S}}\mathbb{E}\left[|u_{1,m}|^2\right], \quad \tau_v = \mathbb{E}\left[|s(\ell)|^2\right], \\ \tau_w &= \frac{\mathbb{E}\left[|\overline{W}_{m,\ell}|^2\right]}{N_{\mathrm{tx}}}, \qquad \alpha \triangleq \frac{\tau_w}{\tau_u\tau_v}. \end{aligned} \tag{16}$$

The spatial second moment denotes the average over array entries. Thus $\|\mathbf{u}_1\|_2^2 = 1$ and unit-power signaling give $\tau_u = \gamma_{\mathrm{S}}/N_{\mathrm{tx}}$, $\tau_v = 1$, and $\tau_w = 1/N_{\mathrm{tx}}$, so $\alpha = 1/\gamma_{\mathrm{S}}$. Here τ_w is the variance of $\mathbf{W}_{\mathrm{r}}$ before the $\sqrt{N_{\mathrm{tx}}}$ factor; the whitened observation noise $\overline{\mathbf{W}}$ still has unit variance. For the fixed-variance asymptotic formulation, equivalently scale $\mathbf{A}$ by $\sqrt{N_{\mathrm{tx}}}$: both τ_u and τ_w are then multiplied by N_{tx}, while α and the singular vectors remain unchanged. The same overlap limit for complex Gaussian noise follows from [16, Sec. 3.1], including deterministic signal directions by unitary invariance. In terms of the factor second moments, the left-factor squared correlation in [15, eq. (5.3)] is

$$\frac{[\beta(\tau_u\tau_v)^2 - \tau_w^2]_+}{\tau_u\tau_v(\beta\tau_u\tau_v + \tau_w)}.$$

Dividing the numerator and denominator by $(\tau_u\tau_v)^2$ and using $\alpha = \tau_w/(\tau_u\tau_v) = 1/\gamma_{\mathrm{S}}$ yields the following limit, used as a prediction for finite N_{tx}, T:

$$\eta \approx \eta_{\mathrm{LM}} \triangleq \frac{[\beta - \alpha^2]_+}{\beta + \alpha}. \tag{17}$$

The informative regime $\gamma_{\mathrm{S}} > \beta^{-1/2}$ concerns asymptotic direction recovery, rather than the false-alarm threshold in (12). In this regime,

$$1 - \eta_{\mathrm{LM}} = \frac{\alpha + \alpha^2}{\beta + \alpha} \sim \frac{1}{\beta\gamma_{\mathrm{S}}} \quad (\gamma_{\mathrm{S}} \to \infty). \tag{18}$$

We use the full expression (17) in the overlap prediction; (18) describes its high-SNR trend. A separate finite-dimensional analysis below establishes how estimation error translates into residual nulling leakage.

C. Residual Leakage Under an Estimated Spatial Null

To isolate estimation-induced leakage, consider a unit-norm desired spatial direction $\mathbf{u}_0$ and a victim whose normalized sensing and downlink signatures are both $\mathbf{u}_1$. This common-signature benchmark does not include frequency-dependent spatial transfer, which is treated in Section IV-B. After aligning the arbitrary common phase of $\widehat{\mathbf{u}}_1$ with $\mathbf{u}_1$, we have $\mathbf{u}_1^{\mathsf{H}}\widehat{\mathbf{u}}_1 = \sqrt{\eta}$. Decomposing $\widehat{\mathbf{u}}_1$ into its projection onto $\mathbf{u}_1$ and an orthogonal residual gives

$$\widehat{\mathbf{u}}_1 = \sqrt{\eta}\,\mathbf{u}_1 + \mathbf{e}, \qquad \mathbf{u}_1^{\mathsf{H}}\mathbf{e} = 0, \qquad \|\mathbf{e}\|_2^2 = 1 - \eta. \tag{19}$$

Define $\rho \triangleq \mathbf{u}_1^{\mathsf{H}}\mathbf{u}_0$ and $\delta \triangleq \mathbf{e}^{\mathsf{H}}\mathbf{u}_0$. Then

$$\hat{\rho} \triangleq \widehat{\mathbf{u}}_1^{\mathsf{H}}\mathbf{u}_0 = \sqrt{\eta}\,\rho + \delta. \tag{20}$$

For $|\hat{\rho}| < 1$, the unit-norm beam that maximizes the desired projection subject to a hard null on the estimated victim direction is

$$\widehat{\mathbf{v}} = \frac{\mathbf{u}_0 - \hat{\rho}\,\widehat{\mathbf{u}}_1}{\sqrt{1 - |\hat{\rho}|^2}}, \qquad \widehat{\mathbf{v}}^{\mathsf{H}}\widehat{\mathbf{u}}_1 = 0. \tag{21}$$

Substituting (19) and (20) into (21) gives

$$\begin{aligned} \widehat{\mathbf{v}}^{\mathsf{H}}\mathbf{u}_1 &= \frac{\mathbf{u}_0^{\mathsf{H}}\mathbf{u}_1 - \hat{\rho}^*\widehat{\mathbf{u}}_1^{\mathsf{H}}\mathbf{u}_1}{\sqrt{1 - |\hat{\rho}|^2}} \\ &= \frac{(1-\eta)\rho^* - \sqrt{\eta}\,\delta^*}{\sqrt{1 - |\sqrt{\eta}\rho + \delta|^2}}. \end{aligned} \tag{22}$$

Under the common-signature benchmark, write the downlink channel in (1) as $\boldsymbol{h}_1 = a_{\mathrm{D}}\mathbf{u}_1$, where $a_{\mathrm{D}} \in \mathbb{C}$ is the overall

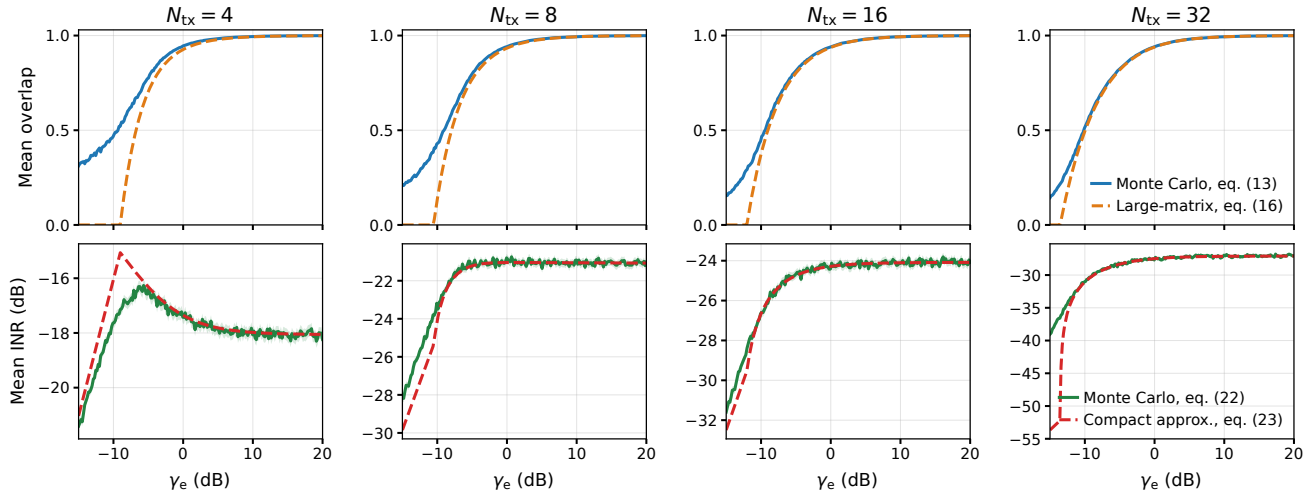


Fig. 3: Mean overlap and residual INR versus per-element sensing SNR $\gamma_e = \gamma_S/N_{tx}$ for a half-wavelength ULA, $T = 16$, $\theta_0 = 0°$, $\theta_1 = 18°$, and $\sigma^2 = 1$. Each point uses 1500 independent trials with block-normalized 16-QAM ($\|\mathbf{s}\|_2^2 = T$), without detection-based selection, and $\gamma_I = \gamma_e$. Dashed curves show the large-matrix overlap and (24) evaluated at η_{LM}. Shading denotes pointwise approximate 95% Monte Carlo confidence intervals for the means. INR is averaged in linear power before conversion to dB.

downlink channel coefficient. Since $\|\mathbf{u}_1\|_2 = 1$, define the downlink interference scale

$$\gamma_I \triangleq \frac{\mathcal{E}_D|a_D|^2}{N_0} = \frac{\mathcal{E}_D\|\boldsymbol{h}_1\|_2^2}{N_0}.$$

Then (1) gives $\mathsf{INR} = \gamma_I|\mathbf{v}^H\mathbf{u}_1|^2$ for any unit-norm beam $\mathbf{v}$. Thus γ_I is the INR attained by a beam aligned with $\mathbf{u}_1$ and is distinct from the uplink sensing SNR γ_S. Without nulling, $\mathbf{v} = \mathbf{u}_0$ gives INR $\gamma_I|\rho|^2$. Under the estimated null, the exact residual INR is

$$\mathsf{INR} = \gamma_I \frac{\left|(1-\eta)\rho^* - \sqrt{\eta}\,\delta^*\right|^2}{1 - \left|\sqrt{\eta}\,\rho + \delta\right|^2}. \tag{23}$$

□

The compact average used to interpret Fig. 3 follows from (23). Condition throughout on η and ρ, and define

$$A = (1-\eta)\rho^* - \sqrt{\eta}\,\delta^*, \qquad x = |\sqrt{\eta}\rho + \delta|^2.$$

For $x < 1$ the denominator admits an expansion; keeping only its leading contribution additionally requires small $|\rho|$ and $|\delta|$, so that $x \ll 1$:

$$\frac{1}{1-x} = 1 + x + O(x^2).$$

Neglecting the higher-order products $|A|^2x$ and above gives $\mathbb{E}[\mathsf{INR} \mid \eta, \rho] \approx \gamma_I\mathbb{E}|A|^2$. Under isotropic error in $\mathbf{u}_1^\perp$, $\mathbb{E}\delta = 0$, and

$$\begin{aligned}\mathbb{E}|A|^2 &= (1-\eta)^2|\rho|^2 + \eta\mathbb{E}|\delta|^2 \\ &\quad - 2(1-\eta)\sqrt{\eta}\,\Re\{\rho^*\mathbb{E}\delta\} \\ &= (1-\eta)^2|\rho|^2 + \eta\mathbb{E}|\delta|^2.\end{aligned}$$

Appendix B derives the conditional projection variance

$$\mathbb{E}|\delta|^2 = \frac{(1-\eta)(1-|\rho|^2)}{N_{tx}-1} \approx \frac{1-\eta}{N_{tx}},$$

where the last step additionally uses small $|\rho|^2$ and a large array. Substitution yields

$$\mathbb{E}[\mathsf{INR} \mid \eta, \rho] \approx \gamma_I\left[(1-\eta)^2|\rho|^2 + \eta\frac{1-\eta}{N_{tx}}\right]. \tag{24}$$

□

The two terms represent coherent mismatch and the residual-error projection into the desired direction, respectively. Figure 3 uses $\eta = \eta_{LM}$ in (24) as a finite-sample prediction; it does not replace the overlap by its high-SNR expansion. The discrepancies at low sensing SNR reflect finite-array fluctuations and the compact leakage approximation. In particular, the finite-dimensional mean overlap approaches $1/N_{tx}$ as the signal vanishes, whereas η_{LM} is zero below its asymptotic threshold. For $N_{tx} = 4$, $|\rho|^2 \simeq 0.249$ also limits the accuracy of the small-correlation approximation in (24).

Finite-dimensional high-SNR behavior: To establish the limiting leakage without taking a large-array limit, fix $N_{tx} > 1$, T, and the unit directions with $|\rho| < 1$, and condition on a waveform with $\|\mathbf{s}\|_2^2 = T$. For the white Gaussian model, Appendix B derives, as $\gamma_S \to \infty$,

$$\mathbb{E}\left[|\widehat{\mathbf{v}}^H\mathbf{u}_1|^2\right] = \frac{1}{T\gamma_S} + o(\gamma_S^{-1}). \tag{25}$$

□

The expectation is over sensing noise. Unlike the compact approximation (24), this leading-order result retains the beam normalization and applies to every fixed nonparallel pair of directions. It is not uniform as $|\rho| \to 1$. For fixed γ_I, the residual INR consequently vanishes. If a common path-loss trend instead produces $\gamma_I = c\gamma_S$ with fixed $c > 0$, then

$$\mathbb{E}[\mathsf{INR}] = \frac{c}{T} + o(1). \tag{26}$$

□

The leakage fraction continues to fall, but the increasing raw interference offsets this improvement. More snapshots reduce the resulting plateau. These conclusions require the common-signature benchmark and fixed block energy; they are not guarantees for arbitrary cross-band channels. In Fig. 3, $\gamma_I = \gamma_e = \gamma_S/N_{tx}$, so $c = 1/N_{tx}$ and the predicted plateau is $1/(N_{tx}T)$.

Controlled validation: All H_1 realizations are included, without detection-based selection. In Fig. 4(a), the normalized mean leakage $T\gamma_S\mathbb{E}[|\widehat{\mathbf{v}}^H\mathbf{u}_1|^2]$ is the ratio of the simulated mean leakage to the leading-order prediction $1/(T\gamma_S)$ in (25). It

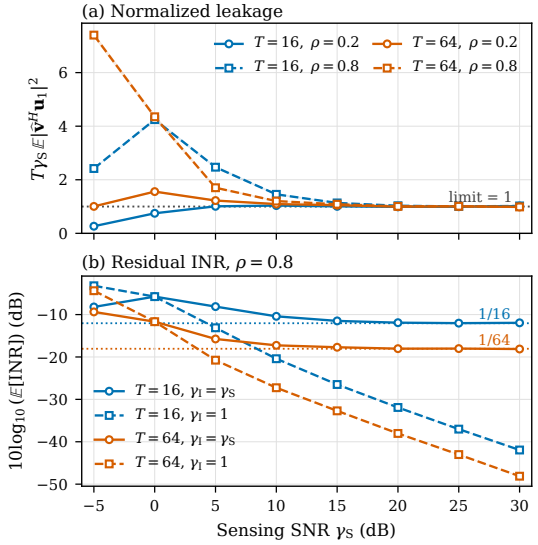


Fig. 4: Finite-dimensional validation with $N_{\mathrm{tx}} = 8$, $T \in \{16, 64\}$, $\|\mathbf{s}\|_2^2 = T$, and 6000 independent noise realizations per SNR. The horizontal axis is γ_S from (13). (a): normalized mean leakage for $|\rho| = 0.2, 0.8$, with high-SNR limit one. (b): mean INR for $|\rho| = 0.8$, comparing $\gamma_I = \gamma_S$ and $\gamma_I = 1$; the coupled limits are $1/T$. Shading denotes pointwise 95% confidence intervals for the Monte Carlo means.

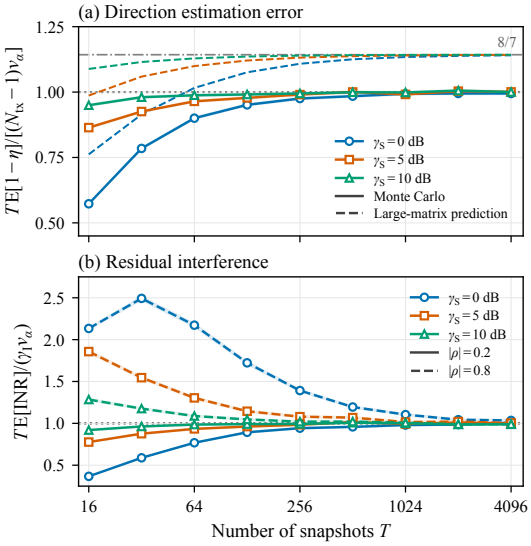


Fig. 5: Long-observation validation with $N_{\mathrm{tx}} = 8$, $\gamma_S \in \{0, 5, 10\}$ dB, $T \in \{16, 32, \ldots, 4096\}$, $\gamma_I = 1$, and $\|\mathbf{s}\|_2^2 = T$. Each (T, γ_S) pair uses 12000 independent white Gaussian noise realizations, all included without detection-based selection. (a): normalized mean overlap loss; solid curves show Monte Carlo means and dashed curves show the large-matrix prediction. The horizontal references are 1 and 8/7. (b): normalized mean residual INR for $|\rho| = 0.2$ (solid) and 0.8 (dashed), with limit 1. Shading denotes pointwise approximate 95% confidence intervals for the Monte Carlo means.

approaches one at high sensing SNR for both tested spatial correlations, while the unnormalized mean leakage tends to zero. In Fig. 4(b), the mean INR approaches $1/T$ when $\gamma_I = \gamma_S$, as predicted by (26), but tends to zero for $\gamma_I = 1$.

D. Long-Observation Behavior at Fixed Sensing SNR

Increasing the snapshot budget provides a second way to improve spatial estimation. Here $\alpha = 1/\gamma_S$ is the inverse sensing SNR, while $\beta = T/N_{\mathrm{tx}}$ is the number of snapshots per array dimension. Interpreting β as measurement time additionally requires an effective snapshot rate and separate accounting for listening gaps and switching overhead.

Fix $\alpha > 0$ and write $\nu_\alpha = \alpha + \alpha^2$. For sufficiently large T, expanding (17) gives

$$1 - \eta_{\mathrm{LM}} = \frac{N_{\mathrm{tx}}\nu_\alpha}{T + N_{\mathrm{tx}}\alpha} = \frac{N_{\mathrm{tx}}\nu_\alpha}{T} + O(T^{-2}). \tag{27}$$

This is an expansion of the large-matrix prediction. The actual fixed-array means follow from a separate perturbation calculation. Retain the common-signature benchmark of Section III-C. Keep $N_{\mathrm{tx}} > 1$, $\gamma_S > 0$, and the unit directions fixed with $|\rho| < 1$, and condition on waveforms of energy $\|\mathbf{s}\|^2 = T$. Under the white Gaussian model, for the unconstrained principal-vector estimator and the hard-null beam (21), Appendix C proves

$$\mathbb{E}[1 - \eta] = \frac{(N_{\mathrm{tx}} - 1)\nu_\alpha}{T} + o(T^{-1}), \tag{28}$$

$$\mathbb{E}[\mathsf{INR}] = \frac{\gamma_I\nu_\alpha}{T} + o(T^{-1}) = \frac{\gamma_I}{T}\left(\frac{1}{\gamma_S} + \frac{1}{\gamma_S^2}\right) + o(T^{-1}), \tag{29}$$

where γ_I is fixed as $T \to \infty$. Expectations include all H_1 realizations, without detection-based selection. The factor $N_{\mathrm{tx}} - 1$ counts the error dimensions orthogonal to the true signature. The leakage coefficient retains the beam normalization and applies to any fixed nonparallel pair of directions. The term α/T comes from signal–noise cross terms; α^2/T comes from fluctuations of the sample noise covariance. Both have the same order in T at fixed SNR.

For $\gamma_I = c\gamma_S$, (29) yields $\mathbb{E}[\mathsf{INR}] = c(1 + 1/\gamma_S)/T + o(T^{-1})$, whose high-SNR coefficient agrees with (26). Thus, longer observations lower the INR plateau, with the mean residual INR vanishing as $T \to \infty$. These asymptotic results describe ensemble means and do not provide deterministic bounds for individual realizations.

Finite-sample validation: Figure 5 tests these limits by increasing T at fixed sensing SNR, complementing the fixed-T, high-SNR experiment in Fig. 4. In Fig. 5(a), the simulated mean overlap loss, normalized by $(N_{\mathrm{tx}} - 1)\nu_\alpha/T$, approaches one. The large-matrix prediction instead approaches $N_{\mathrm{tx}}/(N_{\mathrm{tx}} - 1) = 8/7$ under the same normalization. This finite-array difference distinguishes (27) from (28). Figure 5(b) shows the mean residual INR normalized by $\gamma_I\nu_\alpha/T$. Both tested spatial correlations approach the common unit limit, although the deviations at shorter observation lengths are larger for $|\rho| = 0.8$, particularly at 0 dB sensing SNR. Thus, the leading leakage coefficient is independent of ρ for fixed nonparallel directions, while the finite-T accuracy still depends on their alignment. At 0 dB, $\alpha = \alpha^2 = 1$; retaining only the signal–noise contribution would predict a normalized limit of $1/2$. The results therefore also illustrate why the noise-covariance contribution must be retained in (29) at fixed sensing SNR.

IV. Blind Spatial Sensing and Cross-Band Transfer

The single-victim benchmark isolates spatial estimation error. We now retain anonymous multi-source sensing and explicitly distinguish the monitored uplink carrier f_U from the

protected downlink carrier f_{D}. Sensing first estimates uplink angles and covariance powers; spatial and power transfer then supply the downlink beam design. We then examine how intermittent activity changes the information available to this procedure and the required listening budget.

A. Blind MDL-MUSIC Detection

Throughout this subsection, $\mathbf{u}(\phi,\theta)$ abbreviates the uplink response $\mathbf{u}(f_{\mathrm{U}},\phi,\theta)$, and $G_i = G_i^{\mathrm{U}}$. Using the windows in (3), the BS forms

$$\begin{aligned}\widehat{\mathbf{R}} &= \frac{1}{T}\sum_{\ell=1}^{L_{\mathrm{s}}}\mathbf{Y}_\ell\mathbf{Y}_\ell^{\mathsf{H}} = \sum_{\ell=1}^{L_{\mathrm{s}}}\frac{n_\ell}{T}\widehat{\mathbf{R}}_\ell,\\ \widehat{\mathbf{R}}_\ell &= \frac{1}{n_\ell}\mathbf{Y}_\ell\mathbf{Y}_\ell^{\mathsf{H}}.\end{aligned} \tag{30}$$

Source enumeration, MUSIC, and power fitting all use this pooled covariance. Unequal windows are weighted by their snapshot counts; T in the MDL criterion is the total number of collected snapshots, not the number of windows. All listening windows are retained, including their receiver noise when a source is silent.

For multiple sources represented in the pooled record, the general model (2) applies with unknown source count K, unit-norm steering vectors $\mathbf{u}_i = \mathbf{u}(\phi_i,\theta_i)$, and record-averaged source powers $G_i \triangleq T^{-1}\|\mathbf{b}_i\|_2^2$. The sources need not be active simultaneously in every window. If the empirical cross-correlations $\frac{1}{T}\mathbf{b}_i^{\mathsf{H}}\mathbf{b}_j$ are negligible for $i \neq j$, then the sample covariance obeys

$$\widehat{\mathbf{R}} = \frac{1}{T}\mathbf{Y}\mathbf{Y}^{\mathsf{H}} \approx \sum_{i=1}^{K} G_i\mathbf{u}(\phi_i,\theta_i)\mathbf{u}(\phi_i,\theta_i)^{\mathsf{H}} + \sigma^2\mathbf{I}. \tag{31}$$

Thus, the sensing stage is blind: it does not decode pilots or identify users, but only recovers anonymous direction/power tuples for beam design.

Let $\hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_{N_{\mathrm{tx}}}$ denote the ordered eigenvalues of $\widehat{\mathbf{R}}$. The source count in the pooled record is estimated by the Wax–Kailath MDL rule

$$\hat{K} = \operatorname*{arg\,min}_{0\leq k\leq N_{\mathrm{tx}}-1} \mathrm{MDL}(k), \tag{32}$$

where the explicit criterion is given in Appendix D by (77). The corresponding noise subspace $\mathbf{U}_{\mathrm{n}}$ is formed from the trailing eigenvectors $\hat{\lambda}_{\hat{K}+1},\ldots,\hat{\lambda}_{N_{\mathrm{tx}}}$. We therefore refer to this stage as *Blind MDL-MUSIC Detection*. It then searches for dominant directions:

$$\mathcal{P}_{\mathrm{MU}}(\phi,\theta) = \frac{1}{\mathbf{u}(\phi,\theta)^{\mathsf{H}}\mathbf{U}_{\mathrm{n}}\mathbf{U}_{\mathrm{n}}^{\mathsf{H}}\mathbf{u}(\phi,\theta) + \epsilon_{\mathrm{M}}}, \tag{33}$$

where $\epsilon_{\mathrm{M}} > 0$ is a small regularization constant. The $\hat{K}$ dominant peaks of (33) over the angular grid define the estimated directions $\{(\hat{\phi}_k,\hat{\theta}_k)\}_{k=1}^{\hat{K}}$, and the associated covariance powers are obtained by the covariance-domain nonnegative least-squares fit

$$\{\hat{G}_k\}_{k=1}^{\hat{K}} = \operatorname*{arg\,min}_{\{G_k\geq 0\}} \left\| \widehat{\mathbf{R}} - \hat{\sigma}^2\mathbf{I} - \sum_{k=1}^{\hat{K}} G_k\hat{\mathbf{u}}_k\hat{\mathbf{u}}_k^{\mathsf{H}} \right\|_F^2, \tag{34}$$

where $\hat{\mathbf{u}}_k = \mathbf{u}(\hat{\phi}_k,\hat{\theta}_k)$ and $\hat{\sigma}^2$ is estimated from the smallest eigenvalues of $\widehat{\mathbf{R}}$. The detector therefore outputs the anonymous tuple set $\{(\hat{\phi}_k,\hat{\theta}_k,\hat{G}_k)\}_{k=1}^{\hat{K}}$ at the sensing carrier. The following transfer steps convert these outputs into downlink design inputs.

B. Frequency-Explicit Spatial Transfer

Let f_{U} denote the monitored NTN uplink carrier and f_{D} the TN transmit carrier overlapping the protected NTN receiver. The MDL–MUSIC model above applies at f_{U}: its covariance, noise variance, angular dictionary, and estimated powers are uplink quantities. Introducing frequency labels does not change the stated negligible-cross-correlation assumption, the MDL rule, or the covariance-domain nonnegative gain fit.

For $M = N_{\mathrm{tx}}$ elements at fixed physical coordinates $\mathbf{r}_m$ in meters, use the unit-norm channel-convention response

$$[\mathbf{u}(f,\boldsymbol{\omega})]_m = \frac{1}{\sqrt{M}}\exp\!\left(-\mathrm{j}\frac{2\pi f}{c_0}\mathbf{r}_m^T\mathbf{d}(\boldsymbol{\omega})\right), \tag{35}$$

where $\boldsymbol{\omega} = (\phi,\theta)$ collects the angles, c_0 is propagation speed, and $\mathbf{d}$ points outward from the BS toward the source or departure ray. Array orientation is included in $\mathbf{r}_m$; common element-response amplitudes enter the path coefficient. We use the Hermitian channel convention $y = \mathbf{h}^H\mathbf{w}s + n$. The simulator's raw channel and optimized beam are both conjugated to obtain this convention, leaving received powers unchanged. Sensing, scan dictionaries, and transmit weights must use these conversions consistently; no angular reversal is implied.

For each detected angle $\widehat{\boldsymbol{\omega}}_k$, reconstruct

$$\widehat{\mathbf{u}}_k^{\mathrm{D}} = \mathbf{u}(f_{\mathrm{D}},\widehat{\boldsymbol{\omega}}_k), \qquad \frac{\mathbf{r}_m}{\lambda_{\mathrm{U}}} = \frac{f_{\mathrm{U}}}{f_{\mathrm{D}}}\frac{\mathbf{r}_m}{\lambda_{\mathrm{D}}}, \tag{36}$$

with $\lambda_{\mathrm{U,D}} = c_0/f_{\mathrm{U,D}}$. Thus the physical array is unchanged, including when its spacing is half the reference downlink wavelength. Angular transfer uses estimated geometry and the downlink manifold, without downlink channel truth or instantaneous complex-channel reciprocity [12], [13]. Reusing the uplink vector without reconstruction is a distinct cross-frequency mismatch baseline.

C. Power Normalization and Downlink Penalty Weights

The fitted $\widehat{G}_k^{\mathrm{U}}$ is a covariance power coefficient, not a complex amplitude. For a continuously active source, write $b_i^{\mathrm{U}}(t) = \sqrt{q_i}\,\alpha_i^{\mathrm{U}}s_i(t)$, where $\mathbb{E}|s_i(t)|^2 = 1$ and $q_i = P_i^{\mathrm{U}}/P_{\mathrm{ref}}$. Normalizing received samples by $\sqrt{P_{\mathrm{ref}}}$ gives population signal coefficient $q_i|\alpha_i^{\mathrm{U}}|^2$ and noise variance $N_{\mathrm{BS}}/P_{\mathrm{ref}}$; the empirical definition of G_i^{U} retains finite-record symbol-energy fluctuations. Because steering is unit norm, $|\alpha_i^{\mathrm{U}}|^2$ includes the full array-channel norm for a single spatial mode. The unit-source simulation sets $q_i = 1$; a physical uplink-power interpretation must use the matching noise normalization. Unknown unequal source powers cannot silently be removed from anonymous fitted weights.

Under intermittent transmission, silent snapshots remain in the denominator T, so the fitted powers also reflect each source's activity over the record. They are not estimates of active-transmission power alone. Section IV-D makes this distinction explicit; the frequency correction below does not remove the activity factor.

With equal propagation distance and frequency-independent antenna gains, free-space power gain varies as f^{-2}. The implemented optional compensation therefore forms

$$\widehat{G}_k^{\mathrm{D,nom}} = \kappa \widehat{G}_k^{\mathrm{U}}, \qquad \kappa = \left(\frac{f_{\mathrm{U}}}{f_{\mathrm{D}}}\right)^2, \qquad \mathbf{B}^{\mathrm{D}} = \sum_{k=1}^{\hat{K}} \widehat{G}_k^{\mathrm{D,nom}} \widehat{\mathbf{u}}_k^{\mathrm{D}} (\widehat{\mathbf{u}}_k^{\mathrm{D}})^H. \tag{37}$$

□

Disabling compensation sets $\kappa = 1$. These are nominal penalty weights; unknown source powers, frequency-dependent reflection and antenna gains prevent interpreting them as certified downlink gains. The conversion changes neither detected angles nor the physical channels used for INR/SINR evaluation, and does not control BS power.

For fixed reconstructed directions and detected powers, a common κ is equivalent to replacing the penalty weight λ by $\lambda\kappa$ in Section V. It preserves relative path weights, whereas angular reconstruction changes the penalty's spatial structure. Thus fixed-λ compensation and retuning λ must be distinguished when interpreting comparisons.

Ordinary MDL–MUSIC retains its stated source assumptions. The separate coherent-multipath implementation uses spatial smoothing and a full correlated covariance fit; only its diagonal path-power estimates enter the noncoherent penalty. Transferring these weights does not transfer unknown downlink relative phases or establish coherent-path identifiability for ordinary MUSIC.

D. Intermittent Uplink Activity and Listening Requirements

Activity and accumulated information: Consider $N \geq 1$ protected victims and L_{s} equal-size windows with n_{s} snapshots each, so $T = L_{\mathrm{s}} n_{\mathrm{s}}$. For $i = 1, \ldots, N$, let $A_{i\ell}$ indicate transmission throughout window ℓ. As a tractable activity model, assume $A_{i\ell} \overset{\text{iid}}{\sim} \text{Bernoulli}(p_i)$ across windows. The active-window count is then

$$J_i = \sum_{\ell=1}^{L_{\mathrm{s}}} A_{i\ell} \sim \text{Binomial}(L_{\mathrm{s}}, p_i). \tag{38}$$

Here p_i is the probability of a fully active window, not a detection probability. The window duration and spacing must justify this model; subdividing a correlated burst does not create independent trials. Activities are independent of sensing noise and transmitted symbols, and the spatial signatures remain fixed as in Section II. We hold the protected downlink receiver set fixed regardless of preceding uplink activity. If downlink reception depends on that activity, the count distribution must instead be conditioned on downlink reception.

In window ℓ, write $b_{i\ell}^{\mathrm{U}}(t) = A_{i\ell}\sqrt{q_i}\alpha_i^{\mathrm{U}} s_{i\ell}(t)$ and define $G_{i,\mathrm{act}}^{\mathrm{U}} = q_i |\alpha_i^{\mathrm{U}}|^2$. For fixed active-block energy $\|\mathbf{s}_{i\ell}\|_2^2 = n_{\mathrm{s}}$,

$$G_i^{\mathrm{U}} = \frac{J_i}{L_{\mathrm{s}}} G_{i,\mathrm{act}}^{\mathrm{U}}, \qquad \gamma_{\mathrm{eff},i} = \frac{J_i}{L_{\mathrm{s}}}\gamma_{\mathrm{S},i}, \quad \gamma_{\mathrm{S},i} = \frac{G_{i,\mathrm{act}}^{\mathrm{U}}}{\sigma^2}. \tag{39}$$

For independent unit-variance Gaussian symbols, these are conditional mean powers given J_i; realized block energies also fluctuate. The blind procedure retains every window without using the true $A_{i\ell}$. Thus more collected snapshots need not mean proportionally more signal energy. At fixed λ, less active directions receive smaller nominal penalties even with accurate power estimates. The factor κ does not correct this imbalance, and unknown unequal p_i cannot be absorbed into a single penalty weight.

Connection to the single-victim benchmark: Retain the common-signature, unconstrained principal-vector/hard-null benchmark and assumptions of Section III-D, with fixed active-block energy n_{s}. Given $J = j > 0$, substituting $T = L_{\mathrm{s}} n_{\mathrm{s}}$ and $\gamma_{\mathrm{eff}} = (j/L_{\mathrm{s}})\gamma_{\mathrm{S}}$ into (29) gives

$$\mathbb{E}[\mathsf{INR} \mid J = j] \simeq \gamma_{\mathrm{I}} \left(\frac{\alpha}{j n_{\mathrm{s}}} + \frac{L_{\mathrm{s}}\alpha^2}{j^2 n_{\mathrm{s}}}\right), \qquad \alpha = \gamma_{\mathrm{S}}^{-1}. \tag{40}$$

□

This long-record approximation requires j/L_{s} to approach a positive constant and recovers (29) when $j = L_{\mathrm{s}}$. Silent-window noise adds the factor L_{s}/j in the second term relative to using only jn_{s} snapshots. The approximation excludes $j = 0$ and does not characterize the multi-source MDL–MUSIC beamformer.

Complete INR mixture: For a fixed sensing and beamforming policy, let $F_{i,j}(x) = \mathbb{P}(\mathsf{INR}_i \leq x \mid J_i = j)$ describe the physical downlink INR after the pooled estimate becomes available, and set $\pi_{i,j} = \binom{L_{\mathrm{s}}}{j} p_i^j (1-p_i)^{L_{\mathrm{s}}-j}$. The law of total probability gives the exact relations

$$F_i(x) = \sum_{j=0}^{L_{\mathrm{s}}} \pi_{i,j} F_{i,j}(x), \tag{41}$$

$$\mathbb{P}(\mathsf{INR}_i > \Gamma) = \sum_{j=0}^{L_{\mathrm{s}}} \pi_{i,j}[1 - F_{i,j}(\Gamma)]. \tag{42}$$

The mean INR is likewise $\sum_j \pi_{i,j} m_{i,j}$, where $m_{i,j} = \mathbb{E}[\mathsf{INR}_i \mid J_i = j]$. Each conditional distribution includes estimation error, missed detections, and false alarms under the chosen policy; it is not a point mass at its conditional mean. For the MUSIC pipeline, these distributions require evaluation of the complete detector, transfer, and beamforming procedure. The source-enumeration rule is applied to the pooled record; the window activity model alone does not establish detection performance. The numerical studies in Section VI-D use the calibrated MUSIC implementation specified in Section VI-C. In a multi-source setting, $F_{i,j}$ also depends on other sources' activities and cannot be obtained by reweighting an isolated-victim experiment.

In particular, $\mathbb{P}(J_i = 0) = (1-p_i)^{L_{\mathrm{s}}}$ denotes no uplink evidence from victim i, not necessarily an empty detector output. Other sources or false alarms may still produce directions. If the entire estimated set is empty, $\mathbf{B}^{\mathrm{D}} = \mathbf{0}$ and (49) reduces to desired-link MRT. All such records remain in (41); a detected-only CDF would omit part of the protection risk.

Coverage and listening cost: Let $p_{\min} = \min_{1 \leq i \leq N} p_i$. Without requiring independence between victims, the union bound gives

$$\mathbb{P}(\exists i : J_i = 0) \leq \sum_{i=1}^{N} (1-p_i)^{L_{\mathrm{s}}} \leq N(1-p_{\min})^{L_{\mathrm{s}}}. \tag{43}$$

For $0 < p_{\min} < 1$ and $0 < \epsilon < 1$, hearing every victim at least once with probability at least $1 - \epsilon$ is ensured by

$$L_{\mathrm{s}} \geq \left\lceil \frac{\log(N/\epsilon)}{-\log(1 - p_{\min})} \right\rceil. \tag{44}$$

For small $p_{\min}$, the budget scales as $\log(N/\epsilon)/p_{\min}$. If $p_{\min} = 1$, one window suffices; a victim with $p_i = 0$ cannot be heard from its uplink. Coverage alone guarantees neither accurate detection nor INR protection. Quality-based selection of L_{s} must use (42), including the zero-hit branch.

For the shared-RF arrangement in Fig. 2, let $\tau_{\mathrm{listen}} = n_{\mathrm{s}}/f_{\mathrm{snap}}$ for a specified effective snapshot rate f_{snap}, and let τ_{switch} include both retuning operations per window. With total inter-window gap duration D_{gap} and processing delay τ_{proc},

$$\begin{aligned} C_{\mathrm{RF}} &= L_{\mathrm{s}}(\tau_{\mathrm{listen}} + \tau_{\mathrm{switch}}), \\ t_{\mathrm{ready}} &= C_{\mathrm{RF}} + D_{\mathrm{gap}} + \tau_{\mathrm{proc}}. \end{aligned} \tag{45}$$

The first is RF unavailability; the second is the time until the final beam is usable. To simplify service-time accounting, suppose all BSs follow the same listening and beam-update schedule. Over a static evaluation interval of length H, they serve using MRT during gaps and nonblocking processing, and apply the final beam from t_{ready} onward, with $t_{\mathrm{ready}} < H$. The effective TN spectral efficiency is then

$$R_{\mathrm{eff}} = \frac{D_{\mathrm{gap}} + \tau_{\mathrm{proc}}}{H} R_{\mathrm{MRT}} + \frac{H - t_{\mathrm{ready}}}{H} R_{\mathrm{final}}, \tag{46}$$

where R_{MRT} and R_{final} are the TN spectral efficiencies under MRT and the final nulling beams, respectively, with $R_\bullet = \log_2(1 + \mathsf{SINR}_\bullet)$ for each network beam configuration. The cycle-averaged R_{eff} includes listening and switching overhead. Post-estimation INR in (41) describes the final service phase; risk over all TN transmission time must also include the earlier MRT phases. Listening-induced silence must not be counted as improved nulling. These timing assumptions define a service-cost model. Section VI-D evaluates intermittent activity and final-beam protection; its spatial results do not include this timing protocol's service costs.

V. Noncoherent Downlink Beamforming

The transferred tuples define a spatial penalty for a fixed-power beamforming baseline. Let $\mathbf{h}_0$ be the physical effective channel of the scheduled TN user at f_{D} and define $\mathbf{u}_0^{\mathrm{D}} = \mathbf{h}_0/\|\mathbf{h}_0\|_2$ for $\mathbf{h}_0 \neq 0$. Using the normalized desired direction makes the design weight $\lambda \geq 0$ independent of the desired channel's amplitude. The unit-norm transmit beam solves

$$\max_{\|\mathbf{w}\|_2=1} \; |\mathbf{w}^{\mathsf{H}} \mathbf{u}_0^{\mathrm{D}}|^2 - \lambda \sum_{k=1}^{\hat{K}} \widehat{G}_k^{\mathrm{D,nom}} \left|\mathbf{w}^{\mathsf{H}} \widehat{\mathbf{u}}_k^{\mathrm{D}}\right|^2. \tag{47}$$

Equivalently, define

$$\mathbf{Q} = \mathbf{u}_0^{\mathrm{D}} (\mathbf{u}_0^{\mathrm{D}})^{\mathsf{H}} - \lambda \mathbf{B}^{\mathrm{D}}. \tag{48}$$

The Rayleigh-quotient maximizer is

$$\widehat{\mathbf{w}} = \mathbf{v}_{\max}(\mathbf{Q}). \tag{49}$$

Physical SNR and INR are evaluated using the actual downlink channels and transmit energy in (1), without normalizing away their path gains. Noncoherent here means that the penalty adds directional powers without assigning phases between anonymous paths; it does not assert that actual multipath interference adds incoherently.

A penalty weight trades desired gain against nominal leakage, but does not enforce an INR ceiling or account for sensing time. The spatial performance evaluation below uses this fixed-power tradeoff.

TABLE I: **CDF Simulation Parameters.**

Radio Parameters	**Values**
Downlink Carrier and Bandwidth	8 GHz and 200 MHz
Noise Power Spectral Density	−174 dBm/Hz
BS Transmit Power	35 dBm
Downlink Receiver Noise Power	3.99×10^{-12} W (TN), 1.59×10^{-12} W (NTN)
Multipath order	3 (Reflection and Penetration)
Base Station (BS)	8×8 URA, Directional [18], 3 sectors
Handheld (TN-UE)	(1, 1, 2) Omni-directional [19]
VSAT (NTN-UE)	60 cm diameter, Directional [19]
Antenna Heights	35 m (TN-BS), 1.6 m (outdoor), 1 m (rooftop)
Scene Configuration	
Geographical Location	10 km × 10 km rural area
Radio Material	Wall: Brick, Roof: Concrete, Ground: Medium Dry
Satellite Angle	Azimuth: [0°, 360°], Elevation: [25°, 90°]
Satellite Orbit	550 km (LEO)
NTN-UE Deployment	50 UEs; rooftop probability 0.8
TN-UE Deployment	12 UEs; outdoor sampling probability 0.5
TN-BS Deployment	2×2 layout, 6° downtilt

VI. Simulation Results and Performance Evaluation

A. Multi-Victim Scenario Evaluation

We evaluate the spatial beamforming component through Monte Carlo simulations over a 10 km×10 km rural area near Boulder, Colorado, illustrated in Fig. 6, using the Sionna ray-tracing platform [17]. The terrestrial network consists of four outdoor BS sites on a 2×2 grid, each with three sectors. Each CDF realization contains 50 NTN-UEs and 12 TN-UEs, with one TN-UE sampled for each serving sector subject to the association and TN-link constraints. Table I summarizes the CDF parameters. The common NTN look direction is resampled in each realization with azimuth uniformly distributed over $[0°, 360°]$ and elevation over $[25°, 90°]$. Both $\lambda = 10^{11}$ and $\lambda = 10^{13}$ use the same 20 independently drawn macro realizations, with fixed channels and TN scheduling across the two penalty weights. Each no-nulling sample enters the common baseline once, without repetition across penalty weights or frequencies. The downlink carrier is 8 GHz, and the uplink offsets are −20%, −10%, 0%, +10%, and +20%. For these frequency-offset curves, the spatially smoothed MUSIC detector uses the analytic covariance of the RT channels; finite-snapshot covariance estimation error is not included. The source count is selected using the configured eigenvalue threshold. The CDFs report NTN INR and TN SINR at fixed transmit power.

B. Terrestrial Downlink Interference Nulling

An NTN terminal may receive interference from multiple simultaneously active BS sectors. Let $\mathcal{B}_{\mathrm{act}}$ denote the set of active sectors in one scheduling round. If sector $b \in \mathcal{B}_{\mathrm{act}}$ uses beamformer $\widehat{\mathbf{w}}_b$ and the channel from sector b to NTN

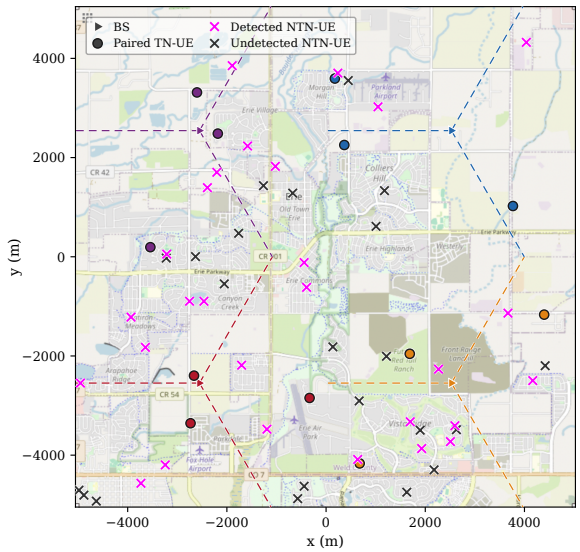


Fig. 6: Representative site-specific layout for the multi-victim nulling evaluation. Triangles mark BS sites, and colored circles mark paired TN users, with colors identifying the serving BS. Magenta and gray crosses indicate detected and undetected NTN terminals, respectively.

terminal i is $\boldsymbol{h}_{i,b}$, then the aggregate downlink INR reported in Figs. 7 and 8(a) is

$$\mathsf{INR}_i^{\mathrm{net}} = \frac{1}{N_0} \sum_{b \in \mathcal{B}_{\mathrm{act}}} \mathcal{E}_{\mathrm{D,b}} \left| \widehat{\mathbf{w}}_b^{\mathsf{H}} \boldsymbol{h}_{i,b} \right|^2 . \tag{50}$$

Figs. 6 and 7 show a separate illustrative deployment with 50 NTN-UEs, 12 paired TN-UEs, 40 m BS heights, and 2° downtilt. The radio maps use uplink and downlink frequencies of 7 GHz and $\lambda = 10^{15}$. Compared with the no-nulling basemap, the proposed nulling suppresses the high-INR regions around the detected NTN terminals.

NTN INR: Fig. 8(a) includes 918 NTN samples with nonzero RT links, including undetected terminals, from the 20 shared drops. The black no-nulling curve contains each sample once; 65.3% of samples have INR at or below −3 dB. At the co-frequency 8 GHz operating point, the post-nulling fractions are 85.2% for $\lambda = 10^{11}$ and 91.9% for $\lambda = 10^{13}$. Across the five uplink frequencies, these fractions range from 82.0% to 85.2% and from 89.5% to 93.1%, respectively. The two penalty weights are compared on identical drops. The gray curve reports the pre-nulling INR of 275 terminals not detected by any sector at the co-frequency operating point, excluding zero-link and zero-INR samples.

The inset in Fig. 8(a) shows that the ±20% offsets generally yield higher residual INR than the corresponding ±10% offsets. Co-frequency sensing is particularly effective for $\lambda = 10^{13}$, whose solid curve yields the lowest INR over most of the magnified region.

TN SINR: Fig. 8(b) contains 240 TN samples for each penalty weight and the same 240 samples in the common no-nulling baseline. The baseline median and fifth-percentile SINR are 22.82 dB and 9.37 dB, respectively. At the co-frequency operating point, the median SINRs are 21.77 dB for $\lambda = 10^{11}$ and 19.76 dB for $\lambda = 10^{13}$, while the fifth-percentile values are 6.96 dB and −7.00 dB. Relative to each user's own no-nulling baseline, 18.3% and 35.4% of TN users

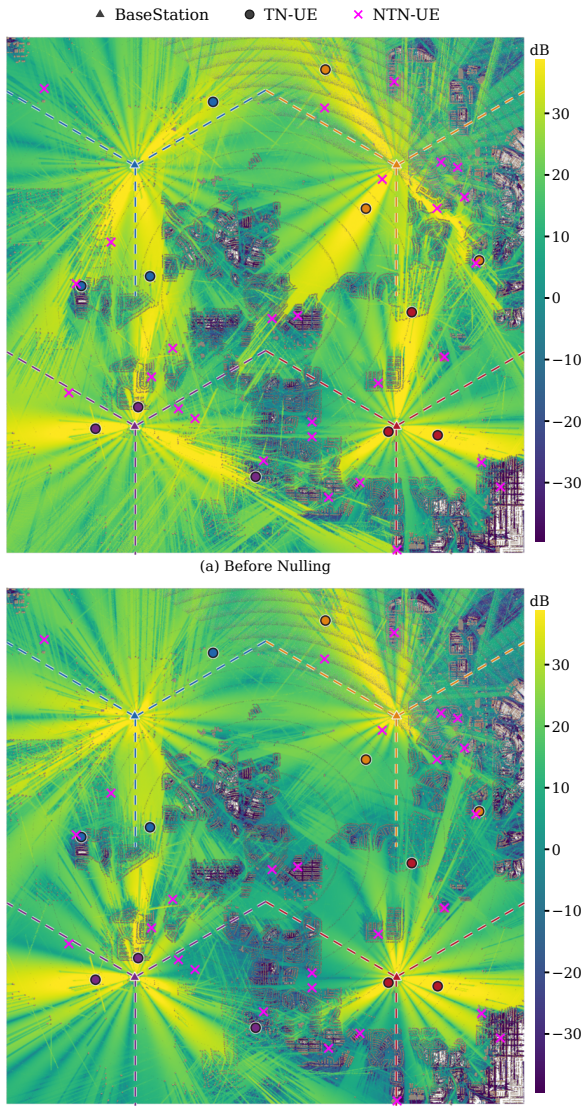
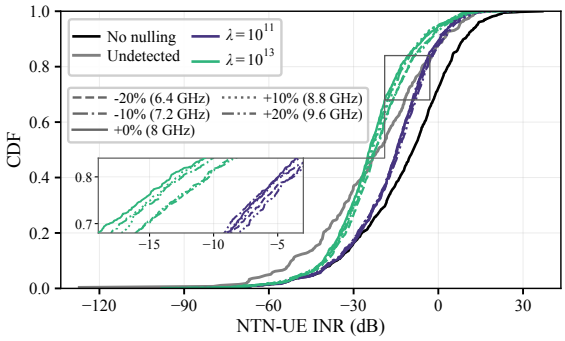


Fig. 7: Downlink INR radio maps before (a) and after (b) applying the proposed nulling method for the deployment in Fig. 6. Triangles, circles, and magenta crosses denote BS sites, paired TN users, and NTN terminals, respectively.

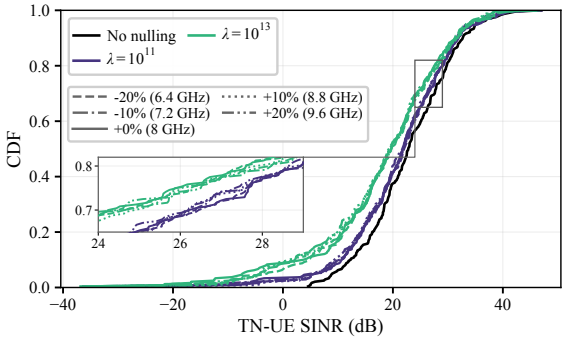
lose more than 3 dB under the two penalty weights. These lower-tail statistics quantify the TN cost alongside the NTN protection benefit.

C. Impact of the Detection Budget

Here, the detection budget is the number T of array snapshots in (3), used to form the sample covariance in (30). We evaluate sensing on the same 20 deployments, with 50 NTN-UEs and 12 active BS sectors per drop. Both carriers are 8 GHz, and $\lambda = 10^{13}$. Channels and TN scheduling remain fixed while T ranges over the eight powers of two from 64 to 8192. Each deployment has five independent waveform/noise repetitions; within a repetition, smaller budgets use prefixes of the same observation record. NTN-UEs transmit continuously, with independent Gaussian source symbols shared across sectors and independent receiver noise. The normalized sensing-noise variance is 3.99×10^{-13}. The spatially smoothed MUSIC implementation uses a T-dependent eigenvalue threshold

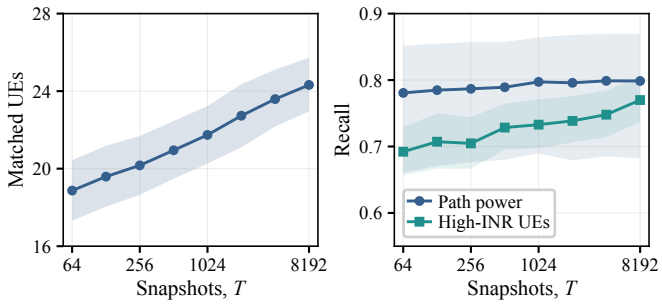


(a) NTN-UE INR



(b) TN-UE SINR

Fig. 8: Empirical CDFs of (a) NTN-UE downlink INR and (b) TN-UE downlink SINR for 50 deployed NTN-UEs per drop. Colors identify λ, and line styles identify uplink offsets from the 8 GHz downlink; solid colored curves use the same uplink frequency. Both penalty weights share 20 independent drops. The black baselines contain 918 NTN and 240 TN samples, each counted once. Gray in (a) shows the pre-nulling INR of 275 undetected terminals with nonzero RT links. Insets magnify the outlined regions using the same colors and line styles. The curves show NTN interference suppression and the associated TN SINR cost.



(a) Detection count (b) Detection coverage

Fig. 9: Finite-snapshot MUSIC detection: (a) mean number of distinct matched NTN-UEs per drop and (b) power-weighted path recall and high-INR UE recall. Shading shows 95% confidence intervals from bootstrap resampling of the 20 deployments, retaining all five repetitions and snapshot budgets together.

calibrated on separate noise-only data in place of the MDL rule in (32). Thus this sweep evaluates the complete detection protocol, including source enumeration, rather than covariance estimation with a fixed threshold.

For offline evaluation, estimated directions are matched one-to-one to RT paths within 5° at each sector. Index the $Q = 20\times5$ deployment–repetition records by q. Let $\mathcal{P}_q$ contain all valid sector–UE–path triples with true ray powers P_{qj},

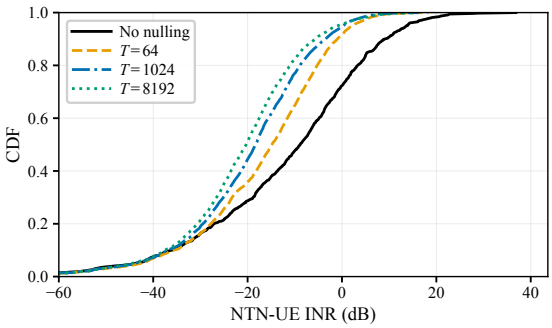


Fig. 10: NTN-UE network INR with finite-snapshot MUSIC at 8 GHz and $\lambda = 10^{13}$. Each nulling curve pools 4590 samples from the same 918 linked terminals across five repetitions, including undetected UEs. The no-nulling baseline counts each terminal once. All samples enter the CDF; the displayed range starts at −60 dB.

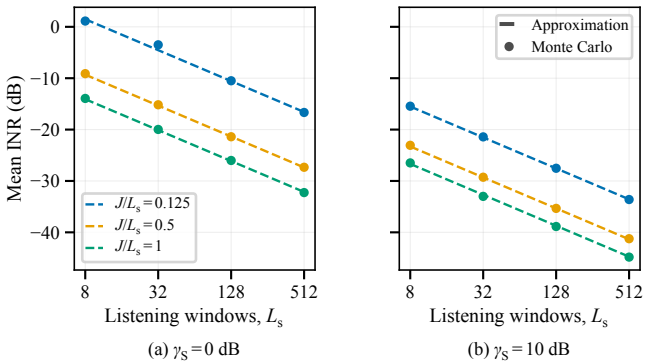
and let $\mathcal{M}_q(T) \subseteq \mathcal{P}_q$ be the matched subset. The set $\mathcal{D}_q(T)$ contains distinct UEs with at least one matched path across the 12 sectors. The fixed set $\mathcal{H}_q$ contains UEs whose no-nulling network INR, evaluated using (50), exceeds −3 dB.

$$\begin{aligned} \overline{N}_{\text{det}}(T) &= \frac{1}{Q}\sum_q |\mathcal{D}_q(T)|, \\ R_{\text{pow}}(T) &= \frac{\sum_q \sum_{j\in\mathcal{M}_q(T)} P_{qj}}{\sum_q \sum_{j\in\mathcal{P}_q} P_{qj}}, \\ R_{\text{high}}(T) &= \frac{\sum_q |\mathcal{D}_q(T)\cap\mathcal{H}_q|}{\sum_q |\mathcal{H}_q|}. \end{aligned} \tag{51}$$

Fig. 9(a) shows $\overline{N}_{\text{det}}$; panel (b) shows R_{pow} (Path power) and R_{high} (High-INR UEs). These are offline coverage scores; RT identities and powers are unavailable to the beamformer. R_{pow} sums individual ray powers without coherent cross terms, while $\mathcal{H}_q$ is unchanged across budgets and repetitions of a deployment. The scores measure detection coverage; Fig. 10 separately measures achieved protection. The activity-coverage bound in (43) concerns missing uplink transmissions; here continuous activity removes silence as a cause of missed detections.

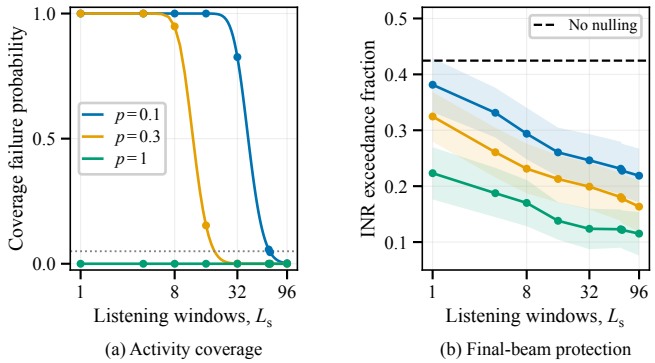
Fig. 9(a) shows that the mean matched count increases from 18.87 at $T = 64$ to 21.74 at $T = 1024$ and 24.32 at $T = 8192$. In Fig. 9(b), power-weighted recall changes from 0.781 to 0.797 and 0.799, while high-INR UE recall rises from 0.692 to 0.733 and 0.770. Much of the individual-ray power is already covered at short observation lengths; the larger budgets provide additional coverage of distinct users, including those exposed to higher baseline interference.

Fig. 10 evaluates the resulting aggregate INR using (50), with the linked UE population held fixed across T. The fractions at or below −3 dB are 86.1%, 90.8%, and 93.0% for $T = 64$, 1024, and 8192, compared with 65.3% without nulling. The corresponding 90th-percentile INRs decrease from −1.01 dB to −3.44 dB and −5.75 dB. These protection gains accompany a TN cost: the fifth-percentile TN SINR decreases from 4.97 dB at $T = 64$ to 1.34 dB at $T = 1024$ and −1.96 dB at $T = 8192$. Detection-budget selection must therefore consider both NTN protection and the TN service loss at the chosen penalty weight.



(a) $\gamma_S = 0$ dB (b) $\gamma_S = 10$ dB

Fig. 11: Conditional single-victim INR with intermittent uplink activity: (a) $\gamma_S = 0$ dB and (b) $\gamma_S = 10$ dB. Dashed curves show (40); markers and error bars show independent Monte Carlo means and 95% confidence intervals, converted from linear INR to dB. Colors indicate J/L_s.



(a) Activity coverage (b) Final-beam protection

Fig. 12: Activity coverage and final-beam protection. (a) Probability that at least one of 50 UEs never transmits: exact independent-activity theory (curves), Monte Carlo estimates (markers), and a 5% target (dotted line). (b) Mean fraction of linked NTN-UEs with final network INR above −6 dB, at 8 GHz and $\lambda = 10^{13}$. Colors identify the same p in both panels. The black dashed line is the no-nulling baseline; shading shows 95% deployment-bootstrap intervals with paired repetitions and budgets.

D. Impact of Intermittent Uplink Activity

We evaluate the conditional INR approximation and activity-coverage requirements of Section IV-D, followed by their implications for the RT network. Each listening window contains $n_s = 64$ snapshots, so $T = 64L_s$. This block size matches the shortest calibrated budget in Section VI-C; it specifies an observation budget rather than a measured traffic-burst duration.

Fig. 11 uses the single-victim principal-vector/hard-null benchmark with eight antennas, $\gamma_I = 10$ dB, desired/victim direction correlation $|\rho| = 0.5$, and fixed active-block energy n_s. For each $L_s \in \{8, 32, 128, 512\}$ and $J/L_s \in \{0.125, 0.5, 1\}$, 2000 independent white-noise realizations are evaluated. Lower activity increases residual INR, while longer records improve the estimate. At $L_s = 512$, all simulated means are within 3% of the approximation. Finite-record deviations remain: at $\gamma_S = 0$ dB, $J/L_s = 0.125$, and $L_s = 32$, the simulated mean is 27% higher. This conditional $J > 0$ benchmark validates the long-record trend; it does not predict the multi-user MUSIC detector.

For the network experiment, we retain the same 20 RT deployments, 50 NTN-UEs, 12 active sectors, and five waveform/noise repetitions per deployment. Both carriers are 8 GHz, $\lambda = 10^{13}$, and $L_s \in \{1, 4, 8, 16, 32, 64, 66, 96\}$. Each UE transmits throughout a window with probability $p \in \{0.1, 0.3, 1\}$, independently across UEs and windows, using Gaussian source symbols; otherwise it remains silent. Channels and scheduling are fixed, and budgets use paired observation prefixes. We use the MUSIC implementation and source-enumeration settings of Section VI-C. All windows enter estimation without knowledge of true activity.

In Fig. 12(a), independence across UEs gives the exact activity-coverage failure probability $1 - [1 - (1 - p)^{L_s}]^{50}$, checked using 10^5 independent trials per operating point. For $p = 0.1$, it decreases from 5.73% at $L_s = 64$ to 4.66% at $L_s = 66$, consistent with the sufficient budget in (44) for $N = 50$ and $\epsilon = 0.05$.

Panel (b) uses a stricter protection threshold of −6 dB, corresponding to a noise-plus-interference increase of $10 \log_{10}(1 + 10^{-6/10}) \simeq 0.97$ dB relative to noise alone. The −3 dB CDF readouts in Figs. 8 and 10 remain separate reference points. We form the indicator that the network INR in (50) exceeds −6 dB, then average over the fixed linked UE population of each deployment, repetitions, and deployments. Never-transmitted and undetected linked UEs remain in this population. The no-nulling fraction is 42.46%. For $p = 0.1$, increasing L_s from 1 to 96 reduces the final exceedance fraction from 38.14% to 21.87%; the corresponding fractions at $L_s = 96$ are 16.34% for $p = 0.3$ and 11.51% for $p = 1$. Activity coverage therefore improves the opportunity for protection, but does not ensure accurate detection or a sufficiently deep null. These results concern the final service phase; listening and switching costs enter separately through (45) and (46).

VII. Conclusion

We connected noncoherent uplink sensing to downlink interference control through three components: a single-victim estimation/leakage benchmark, anonymous multi-source sensing with explicit cross-band transfer, and downlink beam design. Direction reconstruction accounts for the carrier-dependent array manifold, while the nominal power correction preserves a clear distinction between estimated weights and physical downlink gains. The spatial simulations quantify interference suppression and desired-link performance at fixed transmit power. For the common-signature benchmark, the finite-array analyses separate the effects of sensing SNR and observation length. A coupled increase in sensing and interference strength can produce a mean INR plateau at fixed snapshot count, whereas longer records reduce the mean leakage at fixed SNR. These results describe the stated estimation and spatial beamforming models. The intermittent-activity model further separates zero-hit coverage, activity-dependent sensing power, and the full INR mixture, while an explicit listening timeline accounts for the time until the final beam is available. Controlled simulations validate the conditional INR approximation and activity coverage, and the RT study quantifies final-beam protection with intermittent transmissions. Evaluation under measured traffic and hardware timing remains future work.

Appendix A
Profile Likelihood Derivation for the GLRT

This appendix records the algebra behind the profiled GLRT in Section III-A. Under the two hypotheses in (4), the Gaussian likelihoods are

$$p(\mathbf{Y} \mid H_0, \sigma^2) = \frac{1}{(\pi\sigma^2)^{N_{\mathrm{tx}}T}} \exp\left(-\frac{\|\mathbf{Y}\|_F^2}{\sigma^2}\right),$$
$$p(\mathbf{Y} \mid H_1, \mathbf{u}_1, \mathbf{b}, \sigma^2) = \frac{1}{(\pi\sigma^2)^{N_{\mathrm{tx}}T}} \exp\left(-\frac{\|\mathbf{Y} - \mathbf{u}_1\mathbf{b}^{\mathsf{T}}\|_F^2}{\sigma^2}\right). \tag{52}$$

Maximizing the likelihood under H_0 gives

$$\widehat{\sigma}_0^2 = \frac{\|\mathbf{Y}\|_F^2}{N_{\mathrm{tx}}T}. \tag{53}$$

For fixed unit-norm $\mathbf{u}_1$, maximizing the likelihood under H_1 is equivalent to minimizing

$$J(\mathbf{b}) \triangleq \|\mathbf{Y} - \mathbf{u}_1\mathbf{b}^{\mathsf{T}}\|_F^2. \tag{54}$$

Expanding the quadratic form gives

$$\begin{aligned} J(\mathbf{b}) &= \mathrm{Tr}\left\{(\mathbf{Y} - \mathbf{u}_1\mathbf{b}^{\mathsf{T}})(\mathbf{Y} - \mathbf{u}_1\mathbf{b}^{\mathsf{T}})^{\mathsf{H}}\right\} \\ &= \|\mathbf{Y}\|_F^2 - 2\Re\left\{\mathbf{b}^{\mathsf{T}}\mathbf{Y}^{\mathsf{H}}\mathbf{u}_1\right\} + (\mathbf{u}_1^{\mathsf{H}}\mathbf{u}_1)\mathbf{b}^{\mathsf{H}}\mathbf{b}. \end{aligned} \tag{55}$$

Taking the derivative with respect to $\mathbf{b}^*$ and using $\|\mathbf{u}_1\|_2 = 1$ yields

$$\widehat{\mathbf{b}} = \mathbf{Y}^{\mathsf{T}}\mathbf{u}_1^*, \qquad \widehat{\mathbf{b}}^{\mathsf{T}} = \mathbf{u}_1^{\mathsf{H}}\mathbf{Y}, \tag{56}$$

which is the least-squares estimate used to profile out the unknown waveform. Substituting (56) into (54) gives

$$\begin{aligned} \|\mathbf{Y} - \mathbf{u}_1\widehat{\mathbf{b}}^{\mathsf{T}}\|_F^2 &= \|\mathbf{Y} - \mathbf{u}_1\mathbf{u}_1^{\mathsf{H}}\mathbf{Y}\|_F^2 \\ &= \|\mathbf{Y}\|_F^2 - \|\mathbf{u}_1^{\mathsf{H}}\mathbf{Y}\|_2^2, \end{aligned} \tag{57}$$

Therefore, maximizing the profiled likelihood under H_1 reduces to a Rayleigh-quotient problem in $\mathbf{u}_1$, so the ML spatial estimate is the dominant eigenvector in (7).

With $\widehat{\mathbf{u}}_1$ and $\widehat{\mathbf{b}}$ substituted back into the likelihood, the ML estimate of the noise variance under H_1 is

$$\widehat{\sigma}_1^2 = \frac{\|\mathbf{Y}\|_F^2 - \lambda_{\max}(\mathbf{Y}\mathbf{Y}^{\mathsf{H}})}{N_{\mathrm{tx}}T}. \tag{58}$$

Hence the profiled likelihood ratio becomes

$$\Lambda(\mathbf{Y}) = \left(\frac{\widehat{\sigma}_0^2}{\widehat{\sigma}_1^2}\right)^{N_{\mathrm{tx}}T} = \left(\frac{\|\mathbf{Y}\|_F^2}{\|\mathbf{Y}\|_F^2 - \lambda_{\max}(\mathbf{Y}\mathbf{Y}^{\mathsf{H}})}\right)^{N_{\mathrm{tx}}T}. \tag{59}$$

Since $\|\mathbf{Y}\|_F^2 = T\,\mathrm{Tr}(\mathbf{R})$ and $\lambda_{\max}(\mathbf{Y}\mathbf{Y}^{\mathsf{H}}) = T\lambda_{\max}(\mathbf{R})$, (59) is monotone in the normalized largest eigenvalue ξ defined in (11), so thresholding ξ implements the GLRT.

Appendix B
Finite-Dimensional Interpretation of the Large-Matrix Nulling Trend

This appendix separates the compact average model from a fixed-dimensional high-SNR expansion. Let $N_{\mathrm{tx}} > 1$ and $L = |\bar{\mathbf{v}}^{\mathsf{H}}\mathbf{u}_1|^2$. Fix the unit directions $\mathbf{u}_1, \mathbf{u}_0$ independently of the sensing noise. For a fixed waveform, white circular Gaussian noise and the unconstrained principal-eigenvector estimator are invariant under unitary rotations that fix $\mathbf{u}_1$. After phase alignment, this gives conditional isotropy of the error in $\mathbf{u}_1^{\perp}$ and hence

$$\mathbb{E}[\mathbf{e}\mathbf{e}^{\mathsf{H}} \mid \eta, \mathbf{u}_1] = \frac{1-\eta}{N_{\mathrm{tx}} - 1}(\mathbf{I} - \mathbf{u}_1\mathbf{u}_1^{\mathsf{H}}). \tag{60}$$

With the fixed directions and η held fixed, $\mathbb{E}\delta = 0$. Projecting (60) onto $\mathbf{u}_0$ gives

$$\mathbb{E}|\delta|^2 = \frac{(1-\eta)(1-|\rho|^2)}{N_{\mathrm{tx}} - 1}, \tag{61}$$

which is the conditional projection variance used in Section III-C.

For the high-SNR expansion, keep N_{tx}, T fixed, assume $|\rho| < 1$, and fix a waveform with $\|\mathbf{s}\|^2 = T$. Let $\mathbf{P}_{\perp} = \mathbf{I} - \mathbf{u}_1\mathbf{u}_1^{\mathsf{H}}$ and retain the i.i.d. $\mathcal{CN}(0, \sigma^2)$ noise model. As $\gamma_{\mathrm{S}} = |g|^2/\sigma^2 \to \infty$, the phase-aligned principal left singular vector has orthogonal perturbation

$$\mathbf{e} = \frac{\mathbf{P}_{\perp}\mathbf{W}\mathbf{s}^*}{gT} + \mathbf{r}, \qquad \mathbb{E}\|\mathbf{r}\|^2 = O(\gamma_{\mathrm{S}}^{-2}). \tag{62}$$

On $\|\mathbf{W}\|_2 < a|g|\sqrt{T}$ for a sufficiently small fixed $a > 0$, singular-vector perturbation has a quadratic remainder. Fixed-dimensional Gaussian moments and the exponentially small tail probability of the complementary event justify the stated mean-square order. Choosing a small enough also keeps the beam normalization denominator above $(1 - |\rho|^2)/2$ on this event. On its complement, $0 \le L \le 1$ and the Gaussian tail bound make the contribution to the mean leakage exponentially small. These bounds justify averaging the expansions below. Since $\mathbb{E}[(\mathbf{W}\mathbf{s}^*)(\mathbf{W}\mathbf{s}^*)^{\mathsf{H}}] = \sigma^2 T\mathbf{I}$, it follows that

$$\mathbb{E}(1-\eta) = \frac{N_{\mathrm{tx}} - 1}{T\gamma_{\mathrm{S}}} + o(\gamma_{\mathrm{S}}^{-1}), \tag{63}$$

$$\mathbb{E}|\delta|^2 = \frac{1-|\rho|^2}{T\gamma_{\mathrm{S}}} + o(\gamma_{\mathrm{S}}^{-1}). \tag{64}$$

All expectations here are conditional on the fixed waveform and directions. In (23), the numerator's leading term is $|\delta|^2$, while the denominator tends to $1 - |\rho|^2 > 0$. Expanding the normalized projection therefore gives (25): the geometric factor cancels. This calculation does not require small $|\rho|$, but is not uniform as the two directions become parallel.

Finally, multiplying (25) by $\gamma_{\mathrm{I}} = c\gamma_{\mathrm{S}}$, with fixed $c > 0$, proves (26). For random waveform energy, the calculation can instead be conditioned on $S = \|\mathbf{s}\|^2 > 0$, replacing T by S.

Appendix C
Long-Observation Analysis at Fixed Sensing SNR

Fix $N_{\mathrm{tx}} > 1$, $\gamma_{\mathrm{S}} = |g|^2/\sigma^2 > 0$, and unit directions $\mathbf{u}_1, \mathbf{u}_0$ with $q \triangleq 1 - |\rho|^2 > 0$. Condition on a noise-independent

waveform with $\|\mathbf{s}\|^2 = T$ and retain i.i.d. $\mathcal{CN}(0, \sigma^2)$ noise. The estimator is the unconstrained principal left singular vector, phase-aligned as in (19). All expectations include every realization under H_1, without detection-based selection. Write $\mathbf{P}_\perp = \mathbf{I} - \mathbf{u}_1\mathbf{u}_1^{\mathsf{H}}$ and $\nu_\alpha = \alpha + \alpha^2$, where $\alpha = 1/\gamma_{\mathrm{S}}$.

Two first-order perturbations: The population covariance is $\mathbf{R}_* = |g|^2\mathbf{u}_1\mathbf{u}_1^{\mathsf{H}} + \sigma^2\mathbf{I}$, with eigengap $|g|^2$. Expanding the sample covariance gives

$$\begin{aligned}\mathbf{E}_T \triangleq \mathbf{R} - \mathbf{R}_* = \frac{g}{T}\mathbf{u}_1\mathbf{s}^{\mathsf{T}}\mathbf{W}^{\mathsf{H}} + \frac{g^*}{T}\mathbf{W}\mathbf{s}^*\mathbf{u}_1^{\mathsf{H}} \\ + \frac{\mathbf{W}\mathbf{W}^{\mathsf{H}}}{T} - \sigma^2\mathbf{I}.\end{aligned} \tag{65}$$

Projecting the principal-eigenvector equation onto $\mathbf{u}_1^\perp$ and expanding across the fixed eigengap yields

$$\mathbf{e} = \underbrace{\frac{\mathbf{P}_\perp\mathbf{W}\mathbf{s}^*}{gT}}_{\mathbf{a}_T} + \underbrace{\frac{\mathbf{P}_\perp\mathbf{W}\mathbf{W}^{\mathsf{H}}\mathbf{u}_1}{|g|^2T}}_{\mathbf{b}_T} + \mathbf{r}_T. \tag{66}$$

Both displayed perturbations are $O_p(T^{-1/2})$ at fixed SNR; the noise-covariance term cannot be discarded as in a first-order high-SNR expansion.

For noise column $\mathbf{w}_t$, write $z_t = \mathbf{u}_1^{\mathsf{H}}\mathbf{w}_t$ and $\mathbf{v}_t = \mathbf{P}_\perp\mathbf{w}_t$. These are independent proper Gaussian components, with variances σ^2 and $\sigma^2\mathbf{P}_\perp$, respectively; different columns are independent. Since $\mathbf{P}_\perp\mathbf{W}\mathbf{W}^{\mathsf{H}}\mathbf{u}_1 = \sum_t \mathbf{v}_t z_t^*$,

$$\begin{aligned}&\mathbb{E}[\mathbf{a}_T\mathbf{a}_T^{\mathsf{H}}] = \frac{\alpha}{T}\mathbf{P}_\perp, \quad \mathbb{E}[\mathbf{b}_T\mathbf{b}_T^{\mathsf{H}}] = \frac{\alpha^2}{T}\mathbf{P}_\perp, \\ &\mathbb{E}[\mathbf{a}_T\mathbf{b}_T^{\mathsf{H}}] = \mathbf{0}.\end{aligned} \tag{67}$$

The cross moment vanishes by Gaussian third moments; the two perturbations share $\mathbf{W}$ and are not asserted to be independent.

Remainders and leakage: At fixed dimension and positive eigengap, $\mathbb{E}\|\mathbf{E}_T\|_2^4 = O(T^{-2})$. On $\|\mathbf{E}_T\|_2 < a|g|^2$ for sufficiently small fixed a, the eigenvector remainder is quadratic in $\|\mathbf{E}_T\|_2$. The complementary event has exponentially small Gaussian/Wishart tail probability. Bounded eigenvectors and Gaussian moment bounds therefore give $\mathbb{E}\|\mathbf{r}_T\|^2 = O(T^{-2})$. Cauchy–Schwarz bounds its cross moments with the first-order error by $O(T^{-3/2})$, so

$$\mathbb{E}[\mathbf{e}\mathbf{e}^{\mathsf{H}}] = \frac{\nu_\alpha}{T}\mathbf{P}_\perp + o(T^{-1}). \tag{68}$$

Taking the trace proves (28), since $\operatorname{Tr}\mathbf{P}_\perp = N_{\mathrm{tx}} - 1$. Projecting onto $\mathbf{u}_0$ gives $\mathbb{E}|\delta|^2 = q\nu_\alpha/T + o(T^{-1})$. In (23), $1-\eta = O_p(T^{-1})$ and $\delta = O_p(T^{-1/2})$, whereas the denominator tends to q. For $L = |\widehat{\mathbf{v}}^{\mathsf{H}}\mathbf{u}_1|^2$, this gives

$$L = \frac{|\delta|^2}{q} + o_p(T^{-1}), \qquad \mathbb{E}L = \frac{\nu_\alpha}{T} + o(T^{-1}). \tag{69}$$

To justify averaging, choose a still smaller so that the beam normalization denominator exceeds $q/2$ on the good event. Its complement contributes exponentially little because $0 \le L \le 1$; the remaining terms are controlled by the same Gaussian moments. The geometric factor q thus cancels, proving (29). This argument does not require small $|\rho|$, but is not uniform as $|\rho| \to 1$ or $\gamma_{\mathrm{S}} \to 0$.

Relation to the compact prediction: Let m_{c} be (24) evaluated at η_{LM}. Substituting (27) on its informative branch gives exactly

$$\begin{aligned}\frac{m_{\mathrm{c}}}{\gamma_{\mathrm{I}}} &= \frac{\nu_\alpha}{T + N_{\mathrm{tx}}\alpha} + \frac{(N_{\mathrm{tx}}^2|\rho|^2 - N_{\mathrm{tx}})\nu_\alpha^2}{(T + N_{\mathrm{tx}}\alpha)^2} \\ &= \frac{\nu_\alpha}{T} + O(T^{-2}).\end{aligned} \tag{70}$$

The matching leading coefficient follows from canceling the large-matrix overlap factor N_{tx} with the compact projection factor $1/N_{\mathrm{tx}}$. It does not make that prediction exact at finite T.

Fluctuations: Conditioned on all z_t, $\mathbf{a}_T + \mathbf{b}_T$ is proper Gaussian. Its covariance multiplied by T equals $\sigma^2\sum_t |gs_t + z_t|^2\mathbf{P}_\perp/(|g|^4T)$, which converges in probability to $\nu_\alpha\mathbf{P}_\perp$. Together with $\sqrt{T}\mathbf{r}_T \to 0$ in mean square, this yields

$$\begin{aligned}\sqrt{T}\mathbf{e} &\Rightarrow \mathcal{CN}(\mathbf{0}, \nu_\alpha\mathbf{P}_\perp), \\ T(1-\eta) &\Rightarrow \nu_\alpha\,\mathrm{Gamma}(N_{\mathrm{tx}} - 1, 1), \\ TL &\Rightarrow \nu_\alpha\,\mathrm{Exp}(1).\end{aligned} \tag{71}$$

Here Gamma uses shape and scale, and Exp(1) has unit mean. Thus the mean $1/T$ law is not a deterministic protection bound. These distributional limits describe thresholds on the $1/T$ leakage scale, without establishing finite-sample or fixed-threshold large-deviation guarantees.

Appendix D
Blind MDL-MUSIC Detection Justification

This appendix summarizes the uplink detector in (33). Steering vectors and covariance powers are at f_{U}; downlink transfer is applied afterward as in Section IV. For sector t, the received snapshot matrix follows the multi-source model

$$\mathbf{Y}_t = \sum_{k=1}^{K_t} \mathbf{u}(\phi_{k,t}, \theta_{k,t})\mathbf{b}_{k,t}^{\mathsf{T}} + \mathbf{W}_t, \tag{72}$$

where each steering vector is unit norm, the temporal vectors $\{\mathbf{b}_{k,t}\}$ are unknown, and no known pilot structure is assumed. Define the noncoherent source powers

$$G_{k,t} \triangleq \frac{1}{T}\|\mathbf{b}_{k,t}\|_2^2. \tag{73}$$

If the empirical cross-terms $\frac{1}{T}\mathbf{b}_{i,t}^{\mathsf{H}}\mathbf{b}_{j,t}$ are negligible for $i \ne j$, then the sample covariance admits the approximation

$$\widehat{\mathbf{R}}_t \triangleq \frac{1}{T}\mathbf{Y}_t\mathbf{Y}_t^{\mathsf{H}} \approx \sum_{k=1}^{K_t} G_{k,t}\mathbf{u}(\phi_{k,t}, \theta_{k,t})\mathbf{u}(\phi_{k,t}, \theta_{k,t})^{\mathsf{H}} + \sigma_t^2\mathbf{I}, \tag{74}$$

which is the anonymous covariance model used by the detector in Section IV-A.

Let the ordered eigenvalues of $\widehat{\mathbf{R}}_t$ be

$$\hat{\lambda}_{1,t} \ge \hat{\lambda}_{2,t} \ge \cdots \ge \hat{\lambda}_{N_{\mathrm{tx}},t}, \tag{75}$$

with associated eigenvectors $\{\mathbf{q}_{m,t}\}_{m=1}^{N_{\mathrm{tx}}}$. The model order is then estimated by the Wax–Kailath MDL rule

$$\hat{K}_t = \operatorname*{arg\,min}_{0 \le k \le N_{\mathrm{tx}}-1} \mathrm{MDL}_t(k), \tag{76}$$

where

$$\mathrm{MDL}_t(k) = -T\big(N_{\mathrm{tx}} - k\big)\log\left(\frac{\Big(\prod_{m=k+1}^{N_{\mathrm{tx}}}\hat{\lambda}_{m,t}\Big)^{\frac{1}{N_{\mathrm{tx}}-k}}}{\frac{1}{N_{\mathrm{tx}}-k}\sum_{m=k+1}^{N_{\mathrm{tx}}}\hat{\lambda}_{m,t}}\right) + \frac{1}{2}k\big(2N_{\mathrm{tx}} - k\big)\log T. \tag{77}$$

From $\widehat{\mathbf{R}}_t$, the MDL step estimates the signal-subspace dimension $\hat{K}_t$ and yields the decomposition

$$\widehat{\mathbf{R}}_t \approx \mathbf{U}_{\mathrm{s},t}\mathbf{\Lambda}_{\mathrm{s},t}\mathbf{U}_{\mathrm{s},t}^{\mathsf{H}} + \mathbf{U}_{\mathrm{n},t}\mathbf{\Lambda}_{\mathrm{n},t}\mathbf{U}_{\mathrm{n},t}^{\mathsf{H}}, \tag{78}$$

where

$$\mathbf{U}_{\mathrm{s},t} = \big[\mathbf{q}_{1,t},\ldots,\mathbf{q}_{\hat{K}_t,t}\big], \qquad \mathbf{U}_{\mathrm{n},t} = \big[\mathbf{q}_{\hat{K}_t+1,t},\ldots,\mathbf{q}_{N_{\mathrm{tx}},t}\big]. \tag{79}$$

For a candidate steering direction $\mathbf{u}(\phi,\theta)$, the quantity

$$\mathbf{u}(\phi,\theta)^{\mathsf{H}}\mathbf{U}_{\mathrm{n},t}\mathbf{U}_{\mathrm{n},t}^{\mathsf{H}}\mathbf{u}(\phi,\theta) \tag{80}$$

is small when $\mathbf{u}(\phi,\theta)$ lies close to the signal subspace and large otherwise. This directly yields the Blind MDL-MUSIC pseudo-spectrum in (33), whose dominant local maxima provide the anonymous interference directions $\{(\hat{\phi}_{k,t},\hat{\theta}_{k,t})\}_{k=1}^{\hat{K}_t}$.

After the angular peaks are identified, the remaining task is to estimate the noncoherent powers associated with those directions. Let $\hat{\mathbf{u}}_{k,t} = \mathbf{u}(\hat{\phi}_{k,t},\hat{\theta}_{k,t})$ and define the covariance-domain approximation

$$\widehat{\mathbf{R}}_t - \hat{\sigma}_t^2\mathbf{I} \approx \sum_{k=1}^{\hat{K}_t} G_{k,t}\hat{\mathbf{u}}_{k,t}\hat{\mathbf{u}}_{k,t}^{\mathsf{H}}, \tag{81}$$

where $\hat{\sigma}_t^2$ is estimated from the smallest eigenvalues of $\widehat{\mathbf{R}}_t$. A nonnegative least-squares fit of (81) then produces the gains $\{\hat{G}_{k,t}\}$, so the final detector output is the anonymous tuple set $\{(\hat{\phi}_{k,t},\hat{\theta}_{k,t},\hat{G}_{k,t})\}_{k=1}^{\hat{K}_t}$ supplied to the cross-band transfer and noncoherent nulling stages.

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